

ACID Properties

Prepared By: Dr. D. P. Singh

Overview

- **ACID** is a set of properties that ensure database transactions are processed reliably.
- Most SQL databases such as **MySQL, PostgreSQL, and Microsoft SQL Server** follow ACID principles.
- **ACID = Atomicity + Consistency + Isolation + Durability**

Atomicity (All or nothing)

- A transaction is treated as a single unit.
- Either **all operations succeed**. Or **all operations fail** and are rolled backed.
- Example: Transfer Rs. 1000 from account A to B
 1. Debit Rs 1000 from Account A
 2. Credit Rs 1000 in Account B
 - If step 2 fails, step 1 will be roll backed

Consistency

- **Consistency (Valid State to Valid State)**
- A transaction **must preserve all database rules, constraints, and integrity conditions.**
- **Example**
- Suppose a balance cannot be negative:
 - CHECK (balance >= 0)
- Transaction
 - UPDATE Accounts
 - SET balance = -500
 - WHERE id = 1;
- The database rejects the transaction because it violates the constraint.
- **Result:** Database remains consistent.

Isolation

- **Isolation (Transactions Don't Interfere)**
- Multiple transactions running simultaneously should behave as if they were executed one at a time.
- **Example**
 - Two users try to withdraw money simultaneously.
- Without isolation:
 - Balance = 1000
 - User A withdraws 700
 - User B withdraws 500
 - Final balance becomes incorrect.
- With isolation, SQL databases **use locking or concurrency control to prevent such issues.**

Durability

- Once a transaction is committed, the changes survive:
 - System crashes
 - Power failures
 - Server restarts
- After commit, the data is written to durable storage (often through transaction logs).
- Even if the **server crashes immediately afterward, the committed data can be recovered.**

Importance of ACID

- ACID is critical for applications where correctness matters:
 - Banking systems
 - Payment gateways
 - E-commerce orders
 - Airline reservations
 - Inventory management
- These systems cannot tolerate lost, duplicated, or partially completed transactions.

Thank You